

14-1 IEEE 802.11

IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data link layers..

Two LAN technologies are:

1. IEEE 802.11 W/L LAN or W/L Ethernet
2. Bluetooth

Note

A BSS without an AP is called an ad **hoc network;
a BSS with an AP is called an **infrastructure** network.**

Figure 14.1 *Basic service sets (BSSs)*

BSS: Basic service set

AP: Access point

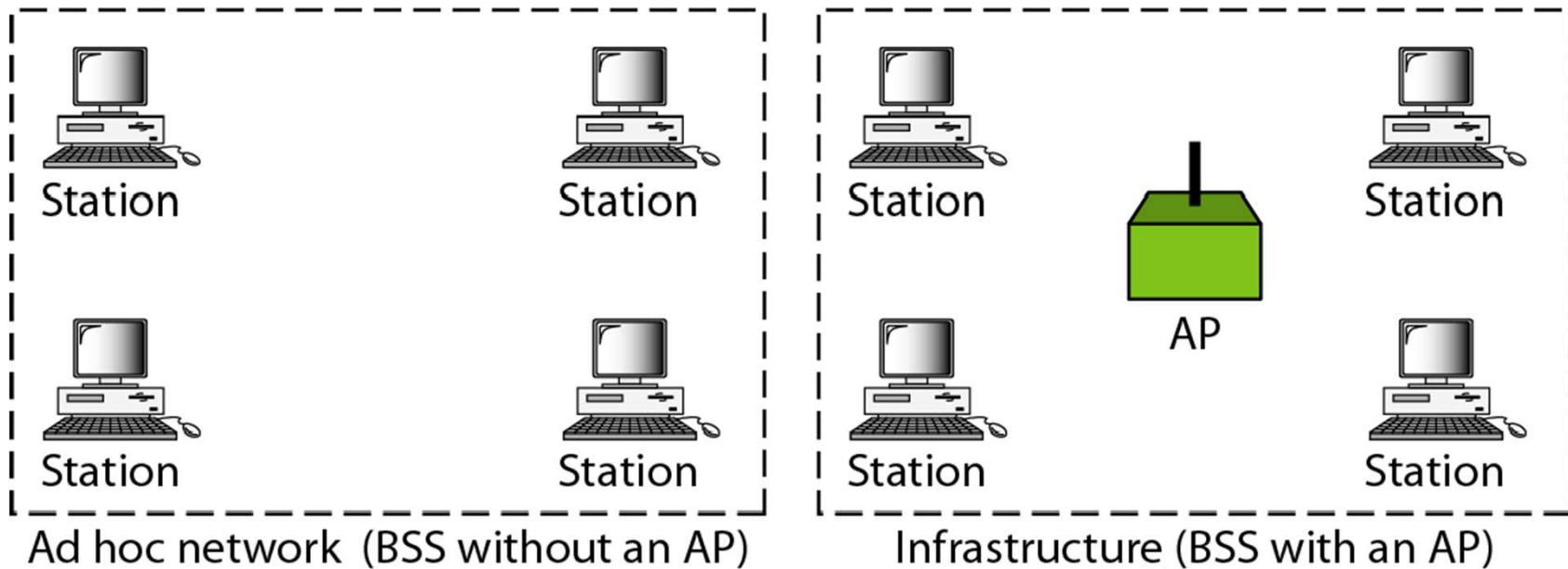


Figure 14.2 *Extended service sets (ESSs)*

ESS: Extended service set
BSS: Basic service set
AP: Access point

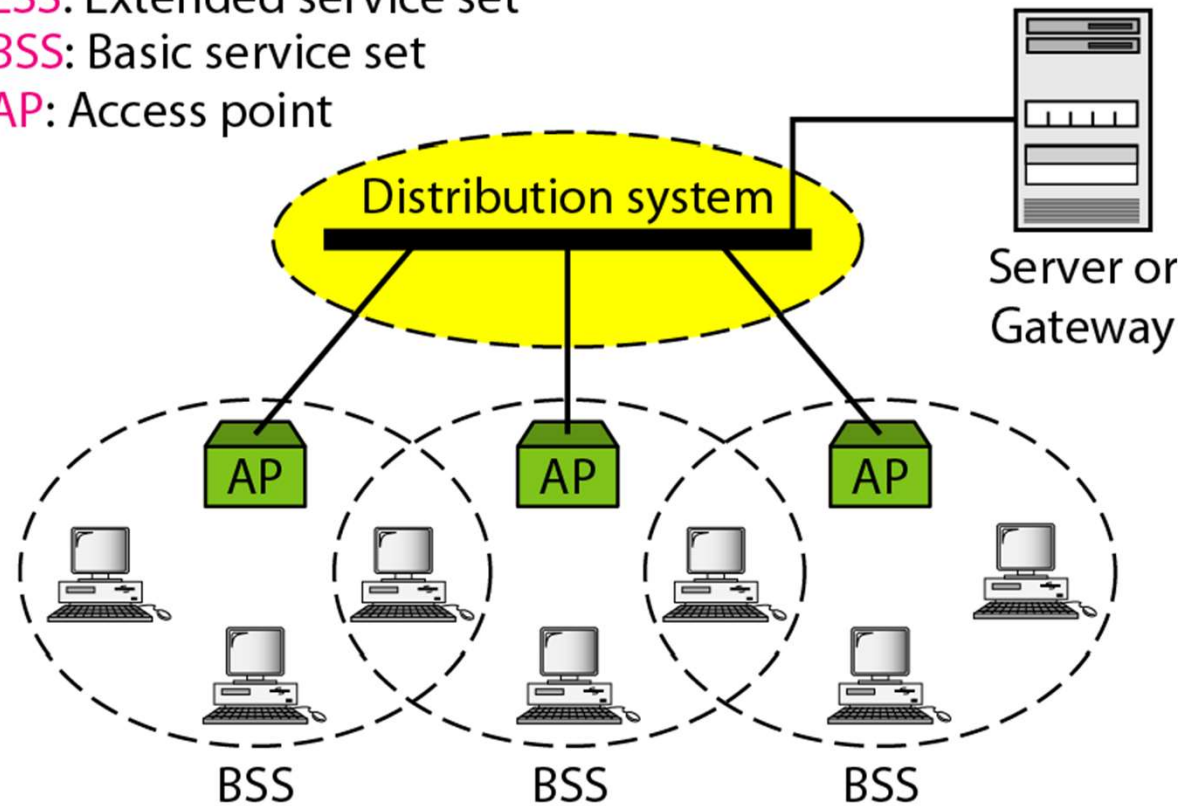


Figure 14.3 *MAC layers in IEEE 802.11 standard*

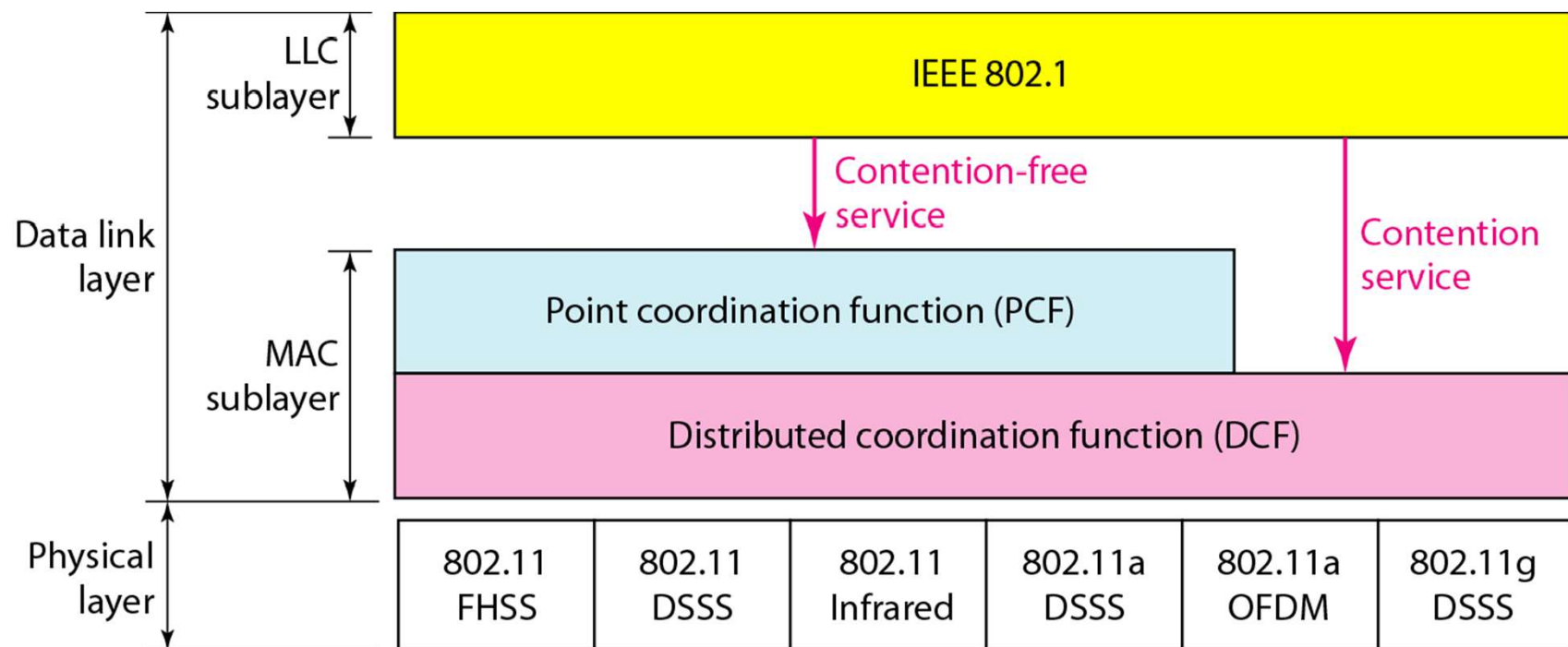


Figure 14.4 *CSMA/CA flowchart*

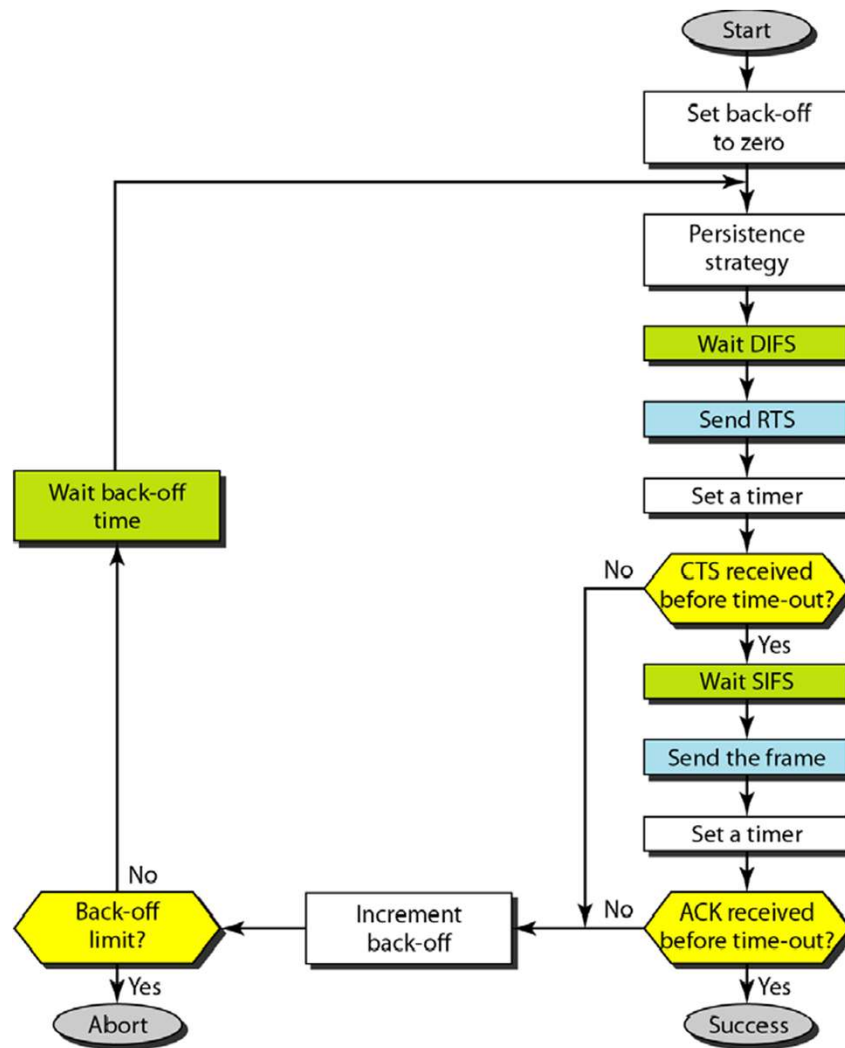
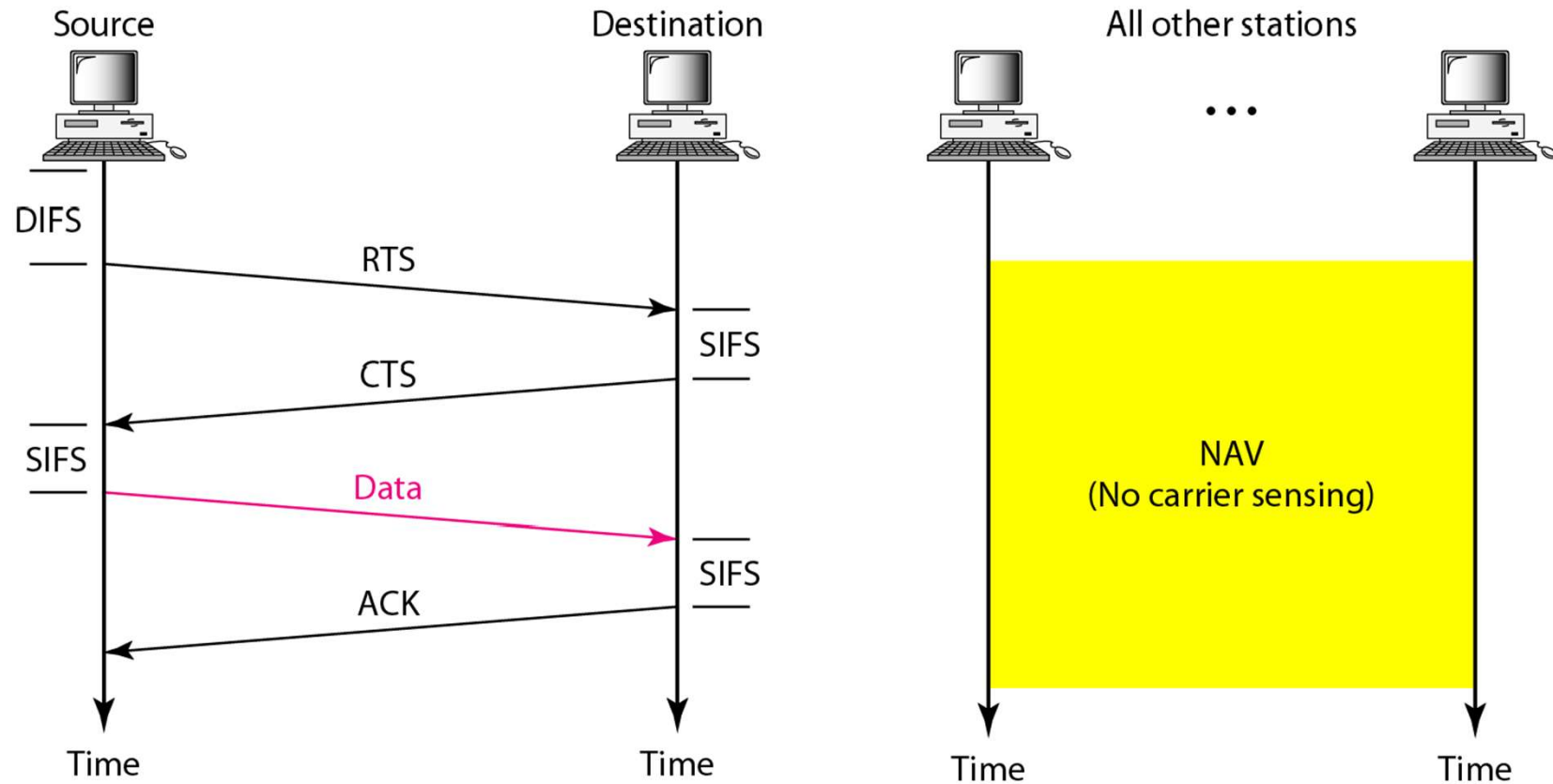


Figure 14.5 *CSMA/CA and NAV*



14-2 BLUETOOTH

***Bluetooth** is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on. A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.*

Figure 14.19 *Piconet*

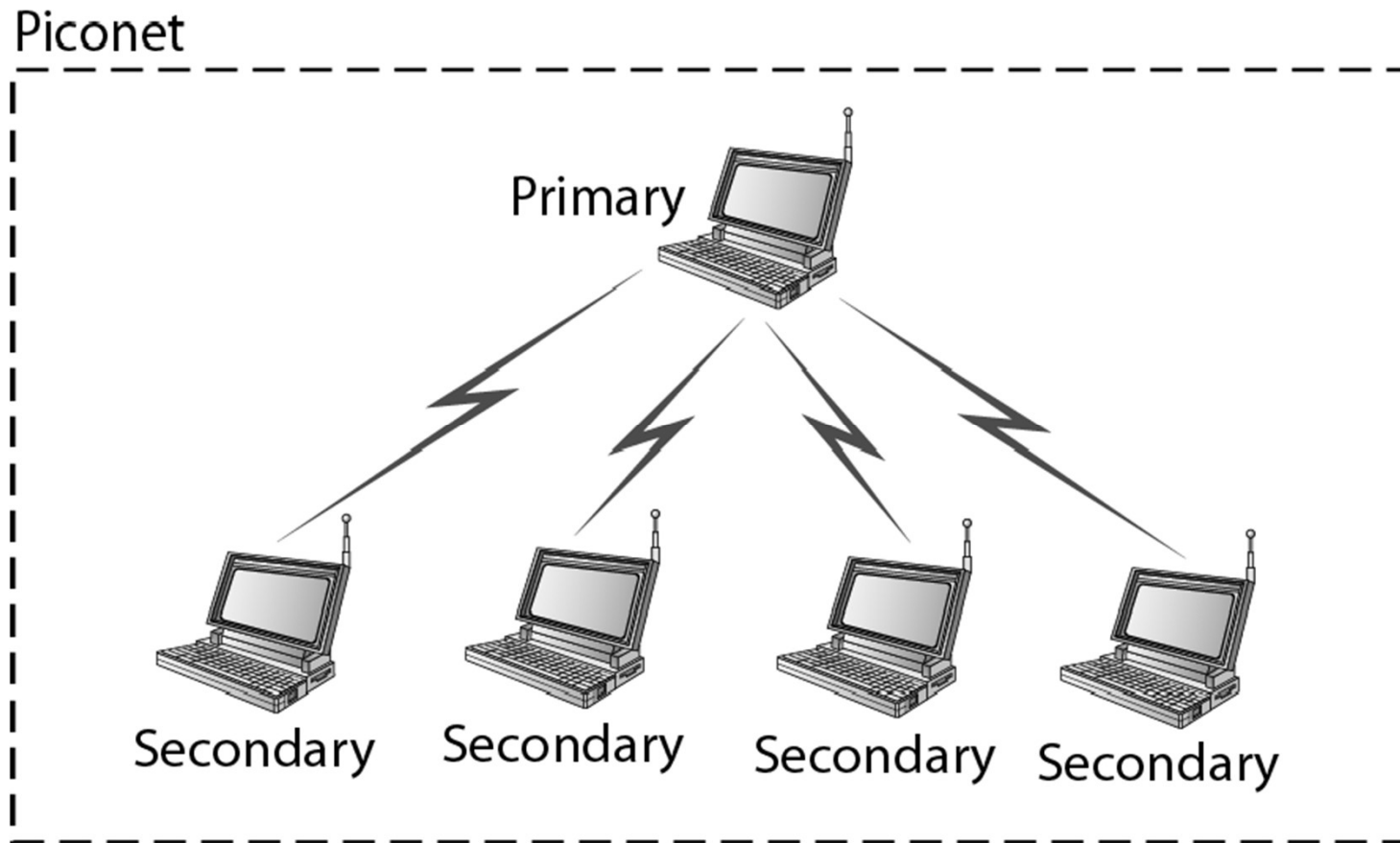


Figure 14.20 *Scatternet*

